Fractional powers

1

a $x^{\frac{1}{2}}$

b $x\frac{1}{6}$

c $x\frac{7}{5}$

d $5\frac{1}{5}$

e $71\frac{1}{7}$

f $5\frac{5}{3}$

g x^{2a}

2 $x^{-}\frac{1}{2}$

3

a $\sqrt{3}$

b $\left(\sqrt[7]{2}\right)^4$

4

a 11	b 1	c 8	d -4
e $\frac{1}{2}$	f 8	g 8	h 5
i 4	j 2	k 216	l 1000
m 4	n 32	o 0	p 256
q $\frac{1}{4}$	r $\frac{1}{9}$	s $2a^2$	t 32

5

a $\frac{3}{8}$	b $\frac{5}{6}$	c $\frac{13}{4}$	d $\frac{8}{9}$
e $\frac{5}{3}$	f $\frac{32}{243}$	g $\frac{3}{4}$	h $243^{\frac{3}{5}}$
i 16	j $2^{\frac{11}{9}}$	k $\frac{x^6}{16}$	

6

a 88

b $8^{-\frac{6}{7}}$

c $\frac{1}{10}$

d $7^{\frac{5}{7}}$

7 4

8

a $25u^8v^6$

b $y^{\frac{10}{3}}$

c $y^{\frac{16}{5}}$

d $4y$

e $2b^{\frac{5}{6}}$

10 It will be easier to use the property $a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m$, finding the square root of 81 (an easily identifiable square number) first and then find the cube of the result. Rather than be required to cube 81 (a large number) and then find the square root of the result.

11

a $\sqrt[4]{x}$

b $\sqrt[10]{x}$

12 $x^{30}y^{18}$

13

a q and r

b r and q

Applications

14 2.56 in

15 600 in^2

16 27 in^3

17 $\$9140$